Python Programming – Problem Statements with Answers

Course: Introduction to Python
Assignment: Practice Problems with Solutions
Total Questions: 10

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1. Even or Odd Number

Program:

```
num = int(input("Enter a number: "))

if num % 2 == 0:
    print("The number is Even")
else:
    print("The number is Odd")
```

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2. Largest of Three Numbers

Program:

```
a = int(input("Enter first number: "))
b = int(input("Enter second number: "))
c = int(input("Enter third number: "))

largest = max(a, b, c)

print("The largest number is:", largest)
```

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3. Factorial of a Number

Program:

```
num = int(input("Enter a number: "))
fact = 1

for i in range(1, num + 1):
    fact = fact * i

print("Factorial =", fact)
```

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4. Fibonacci Series

Program:

```
n = int(input("Enter number of terms: "))

a, b = 0, 1

for i in range(n):
    print(a, end=" ")
    a, b = b, a + b
```

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5. Prime Number Check

Program:

```
num = int(input("Enter a number: "))

if num > 1:
    for i in range(2, num):
        if num % i == 0:
            print("Not a Prime Number")
            break
    else:
        print("Prime Number")
else:
    print("Not a Prime Number")
```

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6. Sum of Digits

Program:

```
num = int(input("Enter a number: "))
sum = 0

while num > 0:
    digit = num % 10
    sum = sum + digit
    num = num // 10

print("Sum of digits:", sum)
```

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7. Reverse a Number

Program:

```
num = int(input("Enter a number: "))
reverse = 0

while num > 0:
    digit = num % 10
    reverse = reverse * 10 + digit
    num = num // 10

print("Reversed number:", reverse)
```

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8. Palindrome Number

Program:

```
num = int(input("Enter a number: "))
temp = num
reverse = 0

while num > 0:
    digit = num % 10
    reverse = reverse * 10 + digit
    num = num // 10

if temp == reverse:
    print("Palindrome Number")
else:
    print("Not a Palindrome Number")
```

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9. Count Vowels in a String

Program:

```
text = input("Enter a string: ")
vowels = "aeiouAEIOU"
count = 0

for char in text:
    if char in vowels:
        count += 1

print("Number of vowels:", count)
```

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10. Simple Calculator

Program:

```
a = float(input("Enter first number: "))
b = float(input("Enter second number: "))

print("1. Addition")
print("2. Subtraction")
print("3. Multiplication")
print("4. Division")

choice = int(input("Choose operation (1-4): "))

if choice == 1:
    print("Result:", a + b)
elif choice == 2:
    print("Result:", a - b)
elif choice == 3:
    print("Result:", a * b)
elif choice == 4:
    print("Result:", a / b)
else:
    print("Invalid choice")
```